

## Module 3

# Measuring Corporate Performance

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FINE 3010, Financial Management  
Prof. Renping Li, Tulane University

# How Do We Evaluate a Corporation's Performance?

- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors

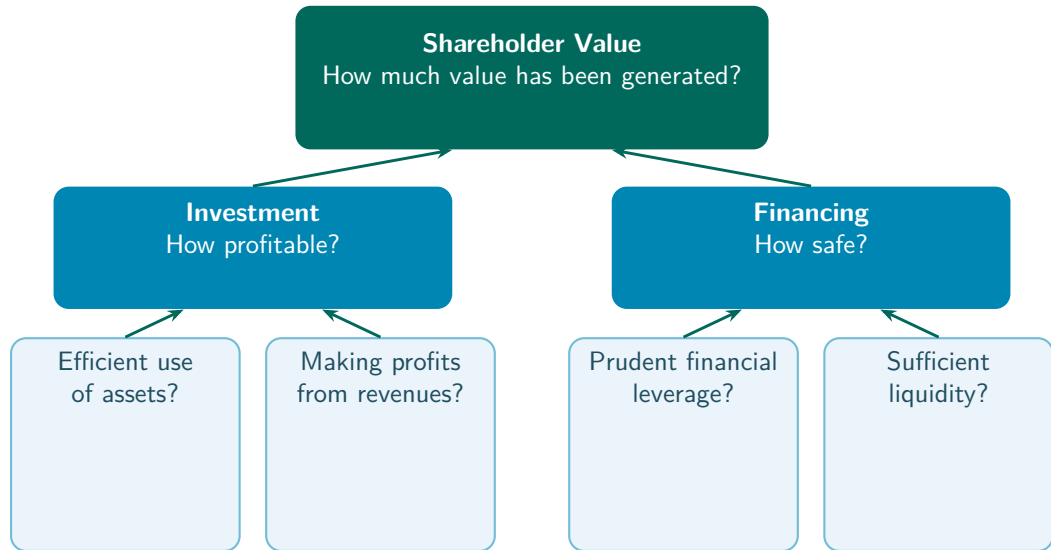


# How Do We Evaluate a Corporation's Performance?

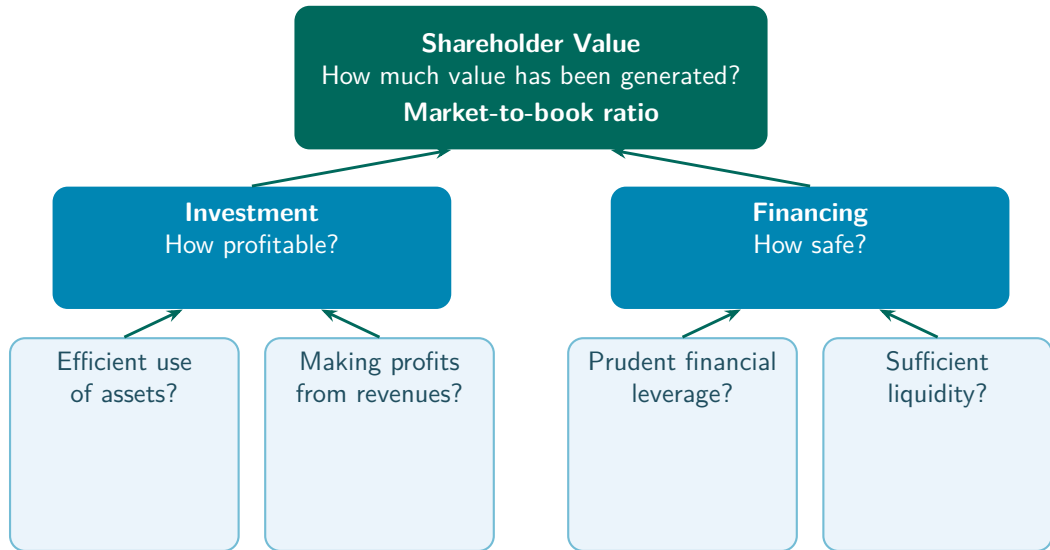
- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors
- Corporate performance is **multi-dimensional**



# Measuring Corporate Performance



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# Shareholder Value

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- **Market capitalization**, another name for the market value of equity

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## Example, Apple

- Recall from **Module 2**, Apple's book value of equity at the end of fiscal year 2025 was \$73,733 Mn
- At the end of 2025, share price about \$271, shares outstanding 14,800 Mn
- Market capitalization =  $\$271 \times 14,800$  Mn = \$4,010,800 Mn



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# Shareholder Value, Market-to-Book Ratio

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## Pros

- Closely tied to how actual investors value the company

## Cons

- Subject to market fluctuations

## Practice Problem

- Here is a simplified balance sheet for a business (figures in \$ millions). It has 600 million shares outstanding with a market price of \$80 a share.

|                     |                 |                                     |                 |
|---------------------|-----------------|-------------------------------------|-----------------|
| Current assets      | \$40,000        | Current liabilities                 | \$30,000        |
| Fixed assets        | 50,000          | Long-term liabilities               | 44,000          |
|                     |                 | Shareholders' equity                | 16,000          |
| <b>Total assets</b> | <b>\$90,000</b> | <b>Total liabilities and equity</b> | <b>\$90,000</b> |

- Calculate the market-to-book ratio.

## Practice Problem

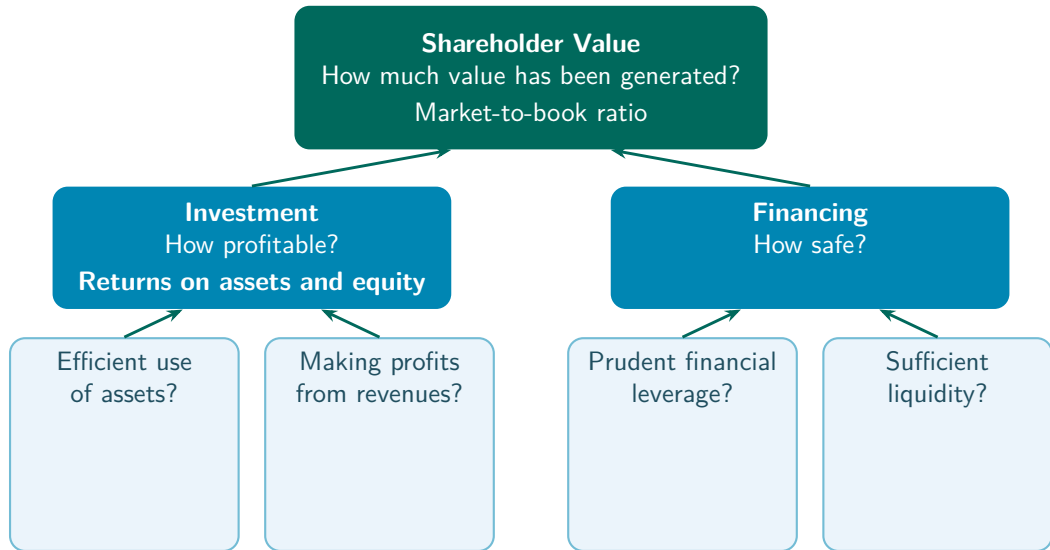
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- Calculate the market-to-book ratio.  $600 \times \$80 / \$16,000 = \$48,000 / \$16,000 = 3.0$



# Measuring Corporate Performance



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## Pros

- Less sensitive to stock market fluctuations

## Cons

- Book values may deviate from market values
- One period, market-to-book reflects all future profits

## Example, Income Statement (\$ Millions)

|                               | \$ Million   |
|-------------------------------|--------------|
| Revenues                      | 80,000       |
| COGS                          | 56,000       |
| SG&A expenses                 | 16,000       |
| Depreciation                  | 3,000        |
| <b>EBIT</b>                   | <b>5,000</b> |
| Interest expense              | 1,000        |
| <b>Taxable income</b>         | <b>4,000</b> |
| Taxes                         | 840          |
| <b>Net income</b>             | <b>3,160</b> |
| Allocation of net income      |              |
| Dividends                     | 1,000        |
| Addition to retained earnings | 2,160        |

## Example, Balance Sheet (\$ Millions)

| Assets                     |               | Liabilities and equity              |               |
|----------------------------|---------------|-------------------------------------|---------------|
| Current assets             |               | Current liabilities                 |               |
| Cash and marketable sec.   | 2,600         | Debt due for repayment              | 200           |
| Accounts receivable        | 500           | Accounts payable                    | 9,900         |
| Inventories                | 9,000         | Total current liabilities           | 10,100        |
| Total current assets       | 12,100        | Long-term liabilities               | 11,300        |
| Fixed assets               |               | Total liabilities                   | 21,400        |
| Tangible fixed assets      |               | Shareholders' equity                |               |
| Property, plant, equipment | 48,200        | Paid-in capital                     | 6,300         |
| Less accum. depreciation   | 19,700        | Retained earnings                   | 13,600        |
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$$\text{Return on equity} = \frac{\text{Net income}}{\text{Shareholders' equity}} = \frac{3,160}{19,900} = 0.159 \text{ (15.9\%)}$$

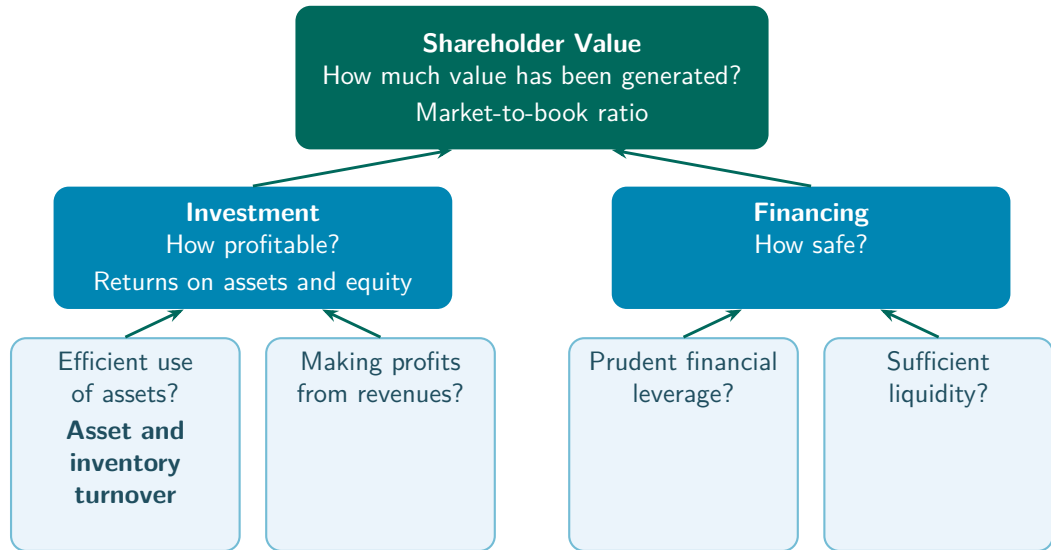
## Practice Problem

- Last year's ROE was 30%. This year the ROE has decreased to 20%, even though the firm's net income equaled last year's. What caused the decrease?
  - a. Shareholders' equity decreased by 10%.
  - b. Shareholders' equity decreased by 50%.
  - c. Shareholders' equity increased by 10%.
  - d. Shareholders' equity increased by 50%.

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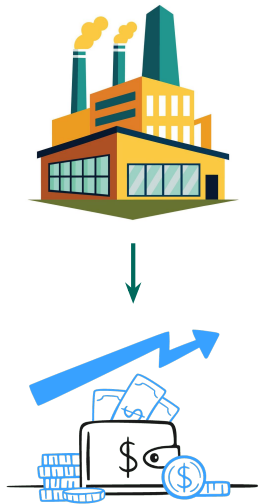
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  - d. Shareholders' equity increased by 50%.
- Suppose net income is  $X$  in both years. Then  $30\% = X / \text{shareholders' equity last year}$ , and  $20\% = X / \text{shareholders' equity this year}$
- Then  $X = 30\% \times \text{shareholders' equity last year} = 20\% \times \text{shareholders' equity this year}$
- So  $\text{shareholders' equity this year} / \text{shareholders' equity last year} = 30\%/20\% = 150\%$ , an increase of 50%

# Measuring Corporate Performance



# Efficiency, Use of Assets

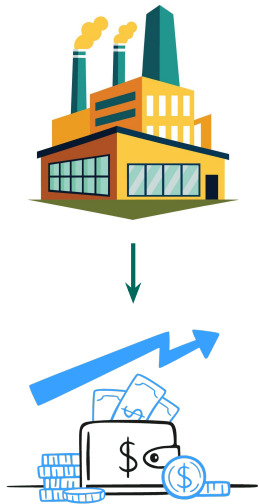
$$\text{Asset turnover ratio} = \frac{\text{Revenues}}{\text{Total assets}}$$



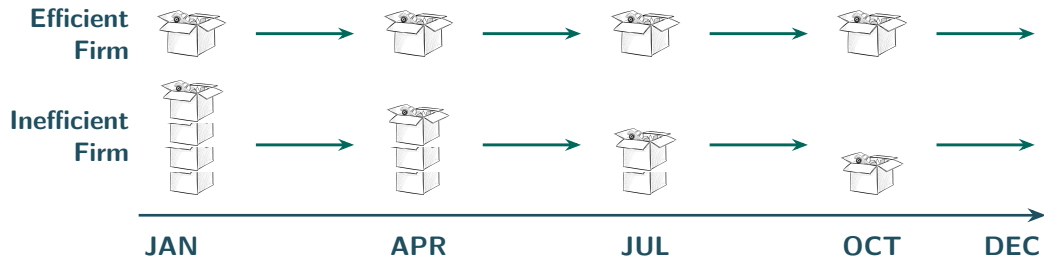
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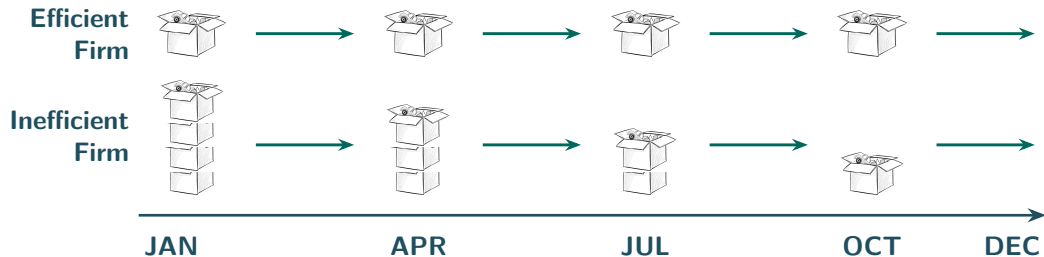
- How much in revenues is generated by each dollar of assets?



# Efficiency, Inventory



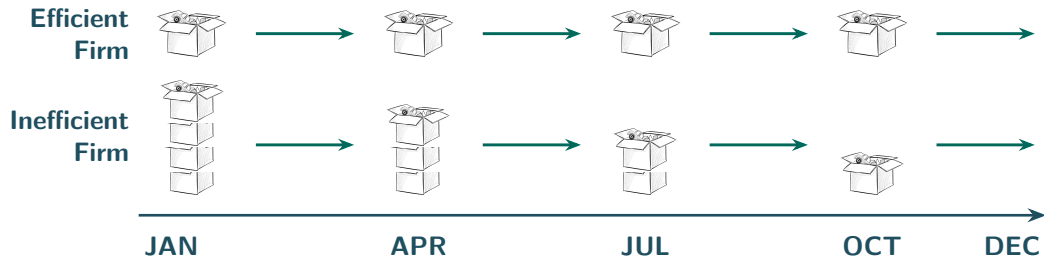
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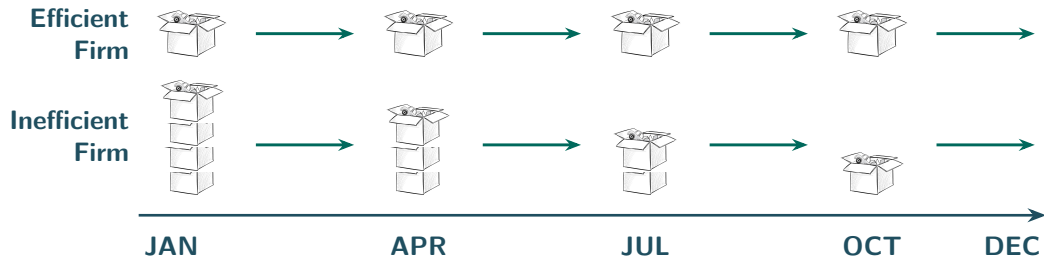
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- Efficient firms do not let inventories sit idle
- Aim for the highest possible? No, inadequate inventories may fail to meet customer demand in time

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- Efficient firms collect bills quickly
- Aim for the fastest possible? No, it may reflect an over-stringent credit policy
  - **Credit policy**, the terms on which a firm lets customers buy now and pay later



## Example, Income Statement (\$ Millions)

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| Revenues                      | 80,000       |
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$$\text{Average collection period} = \frac{500}{80,000/365} = 2.3 \text{ days}$$

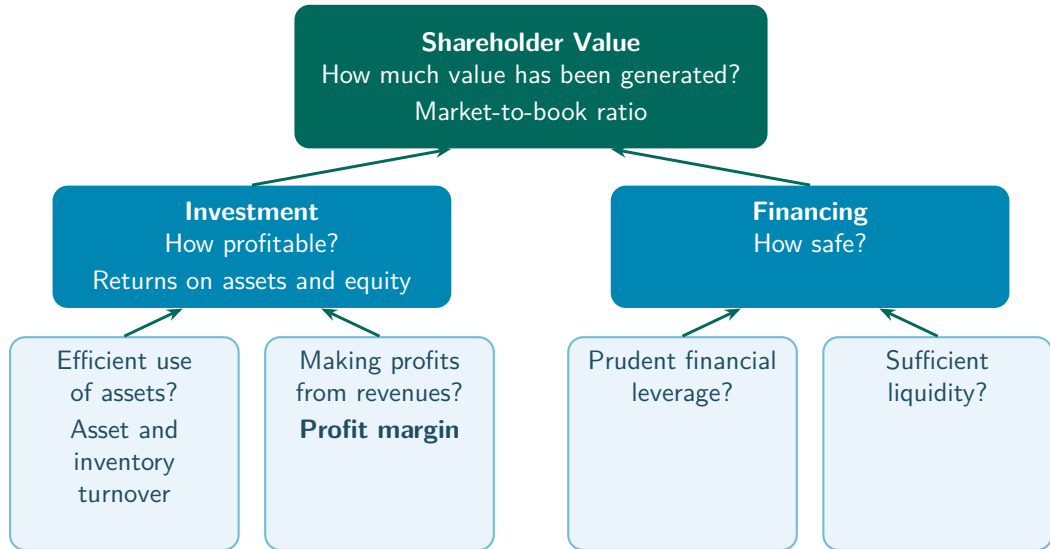
## Practice Problem

- A firm had accounts receivable of \$940 and generated \$3,400 of revenues during the year. How much is its average collection period?
  - a. 94 days
  - b. 95 days
  - c. 101 days
  - d. 107 days

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  - d. 107 days
- Average collection period =  $940 / (3,400 / 365) = 101$  days

# Measuring Corporate Performance





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- How much of each dollar of revenues ends up as net income?

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## Example, Profit Margin

$$\text{Profit margin} = \frac{3,160}{80,000} = 0.0395 \text{ (3.95\%)}$$

# Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?

**Aswath Damodaran**, “Dean of Valuation”



Excess Returns, January 22, 2026  
(1:23), “Could AI Actually Reduce  
Profit Margins? Aswath Damodaran  
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# Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?
- Do you agree?

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$$\text{ROA} = \text{Asset turnover} \times \text{Profit margin}$$

# Decompose ROA

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}} = \frac{3,160}{41,300} = 0.077 \text{ (7.7\%)}$$

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$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}} = \text{Asset turnover} \times \text{Profit margin} = 1.94 \times 0.0395 = 0.077$$

# Decompose ROA

| Company | Strategy       | Asset Turnover | Profit Margin | ROA |
|---------|----------------|----------------|---------------|-----|
| Zara    | Fast fashion   | 1.6            | 7.5%          | 12% |
| Hermès  | Luxury fashion | 0.55           | 22.5%         | 12% |



## Practice Problem

- Which of the following will allow your firm to achieve its targeted 16% ROA with an asset turnover of 2.5?
  - a. A current ratio of 1.6.
  - b. A P/E ratio of 14.
  - c. A return on equity of 25%.
  - d. A profit margin of 6.4%.



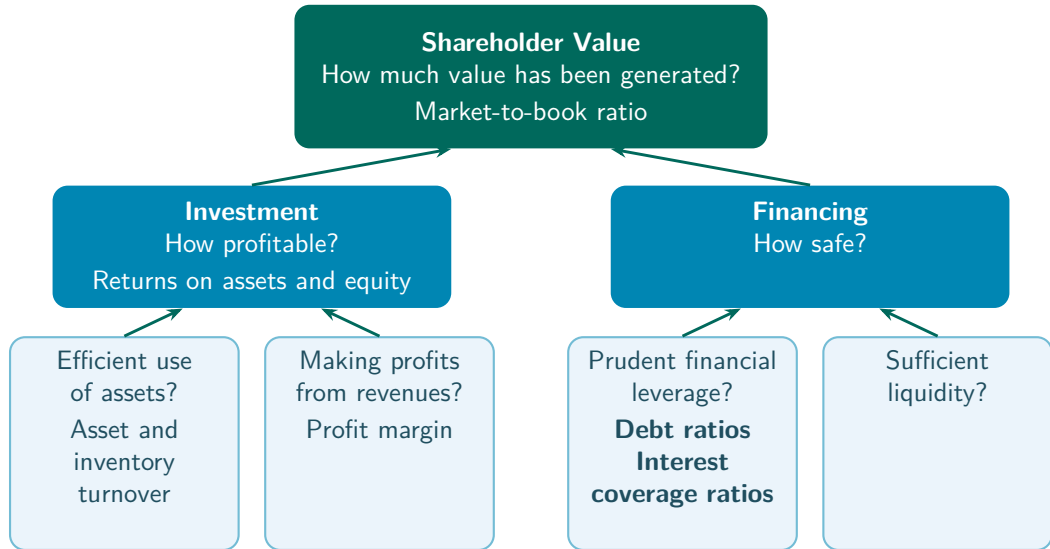
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- $ROA = \text{Asset turnover} \times \text{Profit margin}$ , and  $2.5 \times 6.4\% = 16\%$

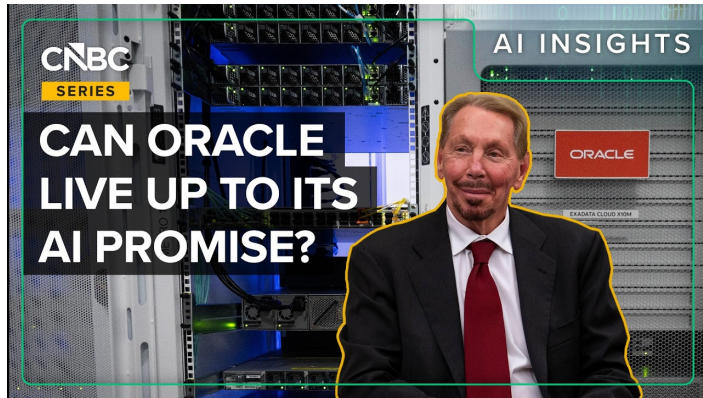
## Recap, Investment

- Investment profitability is measured with the returns on assets and equity
- Efficiency ratios, how much in revenues the assets generate, how quickly inventories sell, and how quickly customers pay
- Profit margin, how much of each dollar of revenues ends up as net income
- $ROA = \text{Asset turnover} \times \text{Profit margin}$

# Measuring Corporate Performance



# Motivating Example, Oracle



CNBC, March 10, 2026 (5:45), “How Oracle’s AI-Fueled Debt Load Has Investors On Edge”

## Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (**good**, 50% chance) or \$2B (**bad**, 50% chance)

## Example, Financing a Data Center

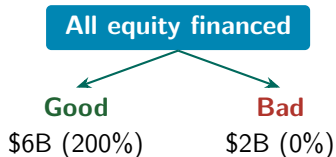
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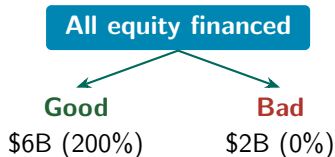
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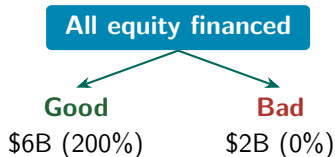
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\$1B equity + \$1B debt

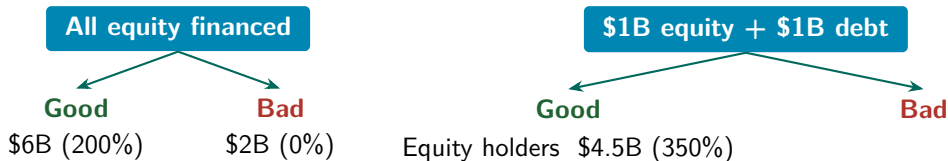
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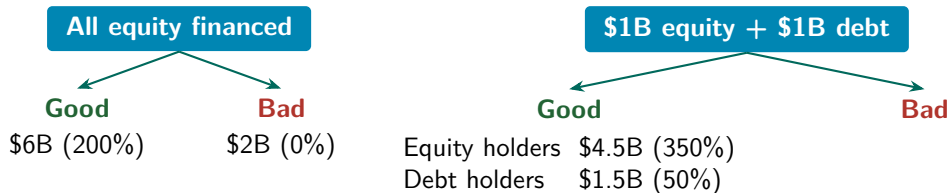
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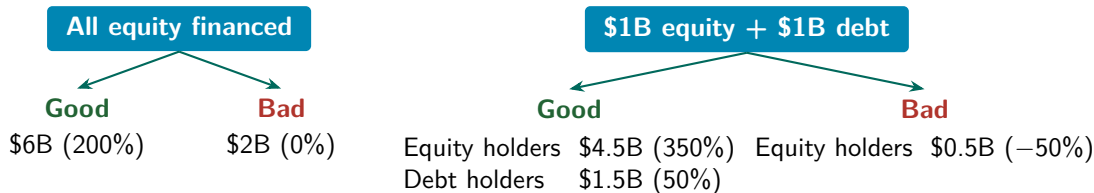
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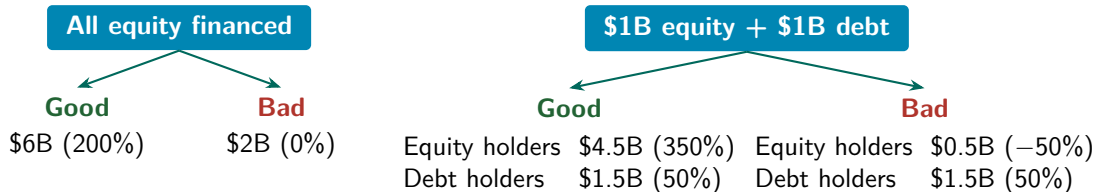
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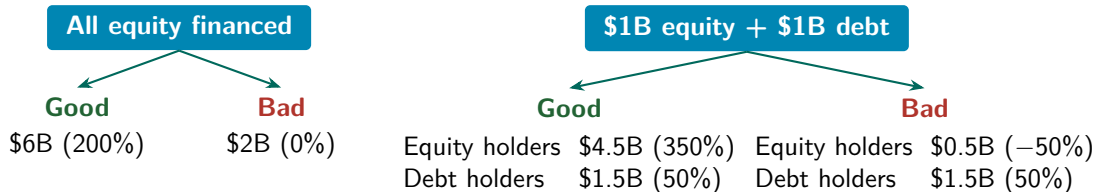
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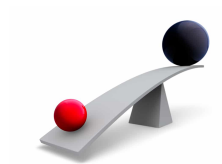
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**With debt, gains and losses to equity holders are amplified**

# Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand





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  - This is **financial leverage**



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- The debt-to-equity ratio is also called the **leverage ratio**
  - It tells to what extent returns to equity investors are amplified because of debt
  - Higher leverage ratio = higher risk of default
    - **Default**, the firm fails to make a required debt payment
    - Default can lead to **bankruptcy**

## Example, Balance Sheet (\$ Millions)

| Assets                     |               | Liabilities and equity              |               |
|----------------------------|---------------|-------------------------------------|---------------|
| Current assets             |               | Current liabilities                 |               |
| Cash and marketable sec.   | 2,600         | Debt due for repayment              | 200           |
| Accounts receivable        | 500           | Accounts payable                    | 9,900         |
| Inventories                | 9,000         | Total current liabilities           | 10,100        |
| Total current assets       | 12,100        | Long-term liabilities               | 11,300        |
| Fixed assets               |               | Total liabilities                   | 21,400        |
| Tangible fixed assets      |               | Shareholders' equity                |               |
| Property, plant, equipment | 48,200        | Paid-in capital                     | 6,300         |
| Less accum. depreciation   | 19,700        | Retained earnings                   | 13,600        |
| Net tangible fixed assets  | 28,500        | Total shareholders' equity          | 19,900        |
| Intangible asset           | 700           |                                     |               |
| Total fixed assets         | 29,200        |                                     |               |
| <b>Total assets</b>        | <b>41,300</b> | <b>Total liabilities and equity</b> | <b>41,300</b> |



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$$\text{Debt ratio} = \frac{\text{Total liabilities}}{\text{Total assets}} = \frac{21,400}{41,300} = 0.52$$

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- Ability of the firm to pay the interest on its debt



## Example, Income Statement (\$ Millions)

|                               | \$ Million   |
|-------------------------------|--------------|
| Revenues                      | 80,000       |
| COGS                          | 56,000       |
| SG&A expenses                 | 16,000       |
| Depreciation                  | 3,000        |
| <b>EBIT</b>                   | <b>5,000</b> |
| Interest expense              | 1,000        |
| <b>Taxable income</b>         | <b>4,000</b> |
| Taxes                         | 840          |
| <b>Net income</b>             | <b>3,160</b> |
| Allocation of net income      |              |
| Dividends                     | 1,000        |
| Addition to retained earnings | 2,160        |

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$$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest expense}} = \frac{5,000 + 3,000}{1,000} = 8.0$$



## Practice Problem

- A firm pays an 8% rate of interest on \$10 million of outstanding debt with face value \$10 million. The firm's EBIT was \$1 million.
  - a. What is its times interest earned?
  - b. If depreciation is \$200,000, what is its cash coverage ratio?

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Interest expense =  $0.08 \times \$10 \text{ million} = \$800,000$   
Times interest earned =  $\$1,000,000 / \$800,000 = 1.25$
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b. If depreciation is \$200,000, what is its cash coverage ratio?

$$\text{Cash coverage ratio} = (\$1,000,000 + \$200,000) / \$800,000 = 1.5$$

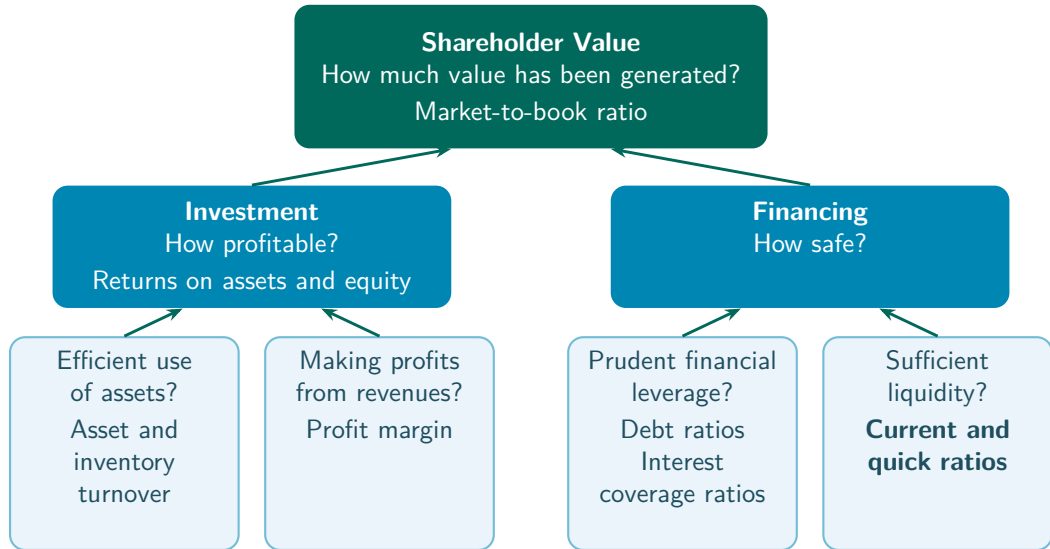
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- Which one of the following will increase a firm's times interest earned ratio?
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  - b. A decrease in cost of goods sold.
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- Lower COGS raises EBIT, the numerator

# Measuring Corporate Performance



# Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value



# Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value
- Liquid assets can be converted into cash quickly and cheaply





## Example, Balance Sheet (\$ Millions)

| Most liquid                                                                       | Assets                     |               | Liabilities and equity              |               | Paid first                                                                          |
|-----------------------------------------------------------------------------------|----------------------------|---------------|-------------------------------------|---------------|-------------------------------------------------------------------------------------|
|                                                                                   |                            |               |                                     |               |                                                                                     |
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|                                                                                   | Net tangible fixed assets  | 28,500        | Total shareholders' equity          | 19,900        |                                                                                     |
| Least liquid                                                                      | Intangible asset           | 700           |                                     |               | Paid last                                                                           |
|                                                                                   | Total fixed assets         | 29,200        |                                     |               |                                                                                     |
|                                                                                   | <b>Total assets</b>        | <b>41,300</b> | <b>Total liabilities and equity</b> | <b>41,300</b> |                                                                                     |

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- Liquidity metrics compare the amount of liquid assets to the amount of debt that requires immediate repayment

## Example, Liquidity

$$\frac{\text{Current assets}}{\text{Current liabilities}} = \frac{12,100}{10,100} = 1.20$$

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$$\frac{\text{Cash} + \text{Marketable securities} + \text{Accounts receivable}}{\text{Current liabilities}} = \frac{2,600 + 500}{10,100} = 0.31$$

## Practice Problem

- Consider this simplified balance sheet for a business.

| Balance Sheet (\$) |              |                     |              |
|--------------------|--------------|---------------------|--------------|
| Current assets     | \$100        | Current liabilities | \$60         |
| Long-term assets   | 500          | Long-term debt      | 280          |
|                    |              | Other liabilities   | 70           |
|                    |              | Equity              | 190          |
| <b>Total</b>       | <b>\$600</b> | <b>Total</b>        | <b>\$600</b> |

- What is the company's debt-to-equity ratio?
- What is its current ratio?

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- What is the company's debt-to-equity ratio?  $(\$60 + 280 + 70)/\$190 = 2.16$
- What is its current ratio?  $\$100/\$60 = 1.67$

## Practice Problem

- A firm uses \$1 million in cash to purchase inventories.
  - a. Will its current ratio rise or fall?
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The current ratio is unaffected. Inventories replace cash, but total current assets are unchanged.
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The quick ratio falls, since inventories are not included in the most liquid assets.

## Recap, Financing

- Financial leverage, debt raises shareholder returns in good times and lowers them in bad times
- Debt-to-equity ratio (the leverage ratio) and debt ratio, how much debt the firm carries
- Times interest earned and cash coverage ratio, the ability to pay the interest on its debt
- Current and quick ratios, do liquid assets cover the debt that requires immediate repayment?

# Key Takeaways

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- Investment, the returns on assets and equity, efficiency ratios, and the profit margin
- Financing, financial leverage, debt and coverage ratios, and liquidity ratios
- Corporate performance is multi-dimensional, no single ratio captures it

# Thank you!

Please let me know if you have any questions.